Some Algebra —

$$M = \begin{bmatrix} a_{11} & \cdots & a_{1m} \\ & & \\ a_{n1} & & a_{nm} \end{bmatrix}_{n\times m}$$

Row Rank = colum rank for $n \times m$ also

↳ over complex numbers

let $\vec{v} \in \mathbb{C}^n$ (complex n dimension)

$M.\vec{v} = \vec{U} \to n \times 1$

↓ rank of M = dimensionality of $\vec{U}$ space

doing linear combination of rows of M and getting $\vec{U}$.

n = row

$$\Rightarrow \vec{U} = \begin{pmatrix} r_1.\vec{v} \\ r_2.\vec{v} \\ \vdots \\ r_n.\vec{v} \end{pmatrix}$$

$r_1 \ldots r_n$ are L.I if $\sum b_i r_i = 0$ iff $b_i = 0 \ \forall i$

$U_i = V_1 a_{i1} + V_2 a_{i2} + \ldots + V_n a_{in}$

$U_i = \sum_{k=1}^{n} V_k a_{ik} = \vec{r}_i \cdot \vec{v}$

$$M = \begin{pmatrix} | & | & | & | \end{pmatrix}$$

↳ first coefficients (a's) → 3rd coefficients (a's)

if they are linearly dependent then — for some V, $U_i = 0$

⇒ rank + nullity = size of matrix

for $\mathbb{F}^2$, rank is even always

⇒ Maple & Sage → powerful symbolic & numeric computing languages.

↳ copy their implementation of diagonalisation

Clifford + T → $C(x)$ ⇒ quadratic always

b/c →

→ into H, T, CZ, & gates i.e. decomposition

for

$$R_\phi = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{bmatrix} \qquad CR_\phi = \begin{bmatrix} 1 & & & 0 \\ & 1 & & \\ & & 1 & \\ 0 & & & e^{i\phi} \end{bmatrix}$$

↓

if Z_8 is done then this is also done.

Example —

let $n = 2$, $|\psi_0\rangle = 0$

i/p variables — 0

let $C(x) = 4x_1x_2 + 4x_2 + x_1 + 7x_3 + 4x_3x_6 + 6x_4 + 4x_4x_6 + 4x_5$

separate Z_8 terms, $C_8(x) = x_1 + 7x_3$ { or take $6x_3$ in C_4 and x_3 in C_8 }

$C(x) = C_8(x) + C_4(x)$

$C_4(x) = 4x_1x_2 + 4x_2 + 4x_3x_6 + 6x_4 + 4x_4x_6 + 4x_5$

opp var → x_5, x_6

$C_4^{\circ}(x) = 4x_1x_2 + 4x_2 + 6x_4$ (→ o/p value)

$C_8(x) = x_1 + 7x_3$

now calculate $\sum_{x \to 1,2,3,4} \left(e^{\frac{i\pi}{4}}\right)^{C_4(x) + C_8(x)}$; $z_0 = e^{i\pi/4}$

for all $x_1 x_3$ combos i.e. $C_8(x)$ values

① $= 1.\sum (z_0)^{4x_2 + 6x_4}$ ($x_1 x_3 = 00$)

$+ z_0 \sum (z_0)^{4x_2 + 4x_4 + 6x_4}$ ($x_1 x_3 = 10$)

$+ z_0^7 \sum (z_0)^{4x_2 + 6x_4}$ ($x_1 x_3 = 01$)

$+ z_0^8 \sum (z_0)^{6x_4}$ ($x_1 x_2 = 11$)

$= 1$

⇒ repeat for all combos of $x_5 x_6$

complexity ⇒ count of these summations

summations are kinda linear/const

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let $C_4^{\circ}(x) = 4x_1x_2 + 4x_2 + 6x_4 + 4x_2x_4$

① becomes ⇒ $1.\sum_{x_2 x_4} (i)^{2x_2 + 2x_4 + 2x_2x_4}$ → mod 4 now

whole expression; 0 or 2; 3 times; 1 times

$(x_1, x_4) \Rightarrow$ 0 → 3 | 1; 1 | 1+2; 2 → 1; 3

Some time this quadratic might become linear after diagonalisation.

Books — Most of things are repetitive so read no. ③ and skim through rest of them.

⇒ Mostly there are theorems so try to do some examples and relate them with what we are doing i.e. polynomials.

① Linear Algebra — K. Hoffman & R Kunze
↳ read chapter 6

② Linear Algebra — Serge Lang
↳ read chapter VIII and IX

③ Adv. Engg. Maths → E. Kreyszig
↳ read section 7.3, 7.4, 7.5, 7.9
chapter 8
section — 9.1, 9.2 and see 20.2 also

④ Math. methods for physicists — Arfken & Weber
↳ read section 3.5